

The following videos provide further information and discussion on soil and vegetation carbon farming through a case study and interviews.

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In this video (8:28 minutes), Nigel Sharp, CEO of Tiverton Agriculture Impact Fund, discusses soil carbon through pasture.

Video: [Soil carbon: Expert Interview with Harry Youngman](#).

My name is Nigel Sharp. I'm CEO of Tiverton Agriculture Impact Fund.

Our main objectives include demonstrating carbon farming and assisting other farmers with their carbon farming ambitions.

Carbon farming is about drawing carbon out of the atmosphere into our soils and growing trees. It's about sequestering carbon, as it's called, the drawdown process. My inspiration for carbon farming comes from the concern of future generations and their livelihoods.

It comes from making profits on the farm and having longevity and resilience in our farming practices.

We have a concern about the longevity of our soils, and so carbon farming helps us think long term about the soil, which helps to build the value of our property. Land appreciation is a key part of our farming productivity.

When we acquired Orana Park, we master planned it so that we could bring changes from conventional agriculture to a carbon farming project. We included shelter belts in our master plan, and we wanted to look at broad acre cropping and olive groves so that we could achieve diversification of income and bring horticulture into the picture. With our shelter belts, we included a sanctuary, which is not that common in farming, but we had some beautiful remnant vegetation that would be home to some fantastic biodiversity, and that could complete our carbon farming picture. Orana was a conventionally fertilised farm at the start. When we acquired it with our master plan, we decided to start using compost, so we were making a lot of compost on the farm.

It was an expensive process, the composting, so we've evolved to a liquid biofertiliser product that we develop on the farm here. The soils have changed since we developed the farming system we've put in place here. The soil structure is stronger, we've got a much higher water holding capacity and a far greater worm count.

As we've evolved the liquid biofertiliser, we've started applying it to all of the crops and the olives here.

In the canola, our oil content is now 52 per cent, which is 10 per cent higher than the district average. So we're seeing, significant changes to the health of the produce, and we're seeing the soil starting to sequester carbon.

Orana Park has been master planned over a long term. So in putting in a thousand hectares of olives, they take seven years to develop. We started the changes of composting moving to liquid biofertiliser.

So the profitability is a steady grow that then hopefully ends up in an exponential type curve. So we're expecting in another three years, which is seven years into ownership of this property, to be highly profitable, much more significantly profitable than when we commenced. The resilience in having a more resilient cropping regime as a result of carbon farming is really critical because it can extend the season in a dry season. It can assist us in a wet season. So it makes the farming, more adaptable to climate change and varying weather conditions that we're experiencing.

We've registered a soil carbon project here. When we started the soil carbon project, there's 0.7 per cent carbon in the soil. We're hoping to take that to 2.7. That should lead to carbon credits and carbon profitability for us.

Also, in developing our shelter belts and setting up the sanctuary, we're looking forward to biodiversity credits becoming a market. Asset value is critical to farmers. Carbon farming is extending the life of our soils, hopefully, in perpetuity, and that it's a productivity of a farm that ultimately drives the value of your asset. We believe carbon farming is going to increase our asset above the historical asset growth targets.

The measurements are guided by the Clean Energy Regulator, we're also using some satellite technology to help us monitor the changes to our soil and the carbon content, and to help us in making predictions and our planning, with the use of our interventions such as the liquid biofertiliser.

Being able to get, almost on time measurement, means that we can have a look at the changes, and the applications that we might make even within one calendar year. There's a significant process to register a soil carbon project. There's completing the application. There's doing a farm plan. There's baselining and collecting all the samples so that the baseline can be registered as well. So, again, we're coming back to talk to other farmers that have been through this process to make it easier because when you just start reading all of the information required, it can feel overwhelming.

When we acquired Orana Park, we set up a master plan, which included shelter belts and biodiversity corridors throughout the entire farm. Those plantings qualify for carbon credits through the REMP, part of the Emissions Reduction Fund. We're intending to complete those plantings, and they become part of our carbon farming story. You can have soil carbon and tree carbon projects on your property.

There's not a double count where soil and trees are grown on the same piece of land. That is a tree carbon project. The soil carbon project will sit separately from where you are growing trees. It's important to be aware of the risks when you transition to carbon farming.

It may be such as changing some of your machinery or equipment. In our experience, it was putting a liquid injection system into our seeder. It can be owning additional tanks. Some of the other transitional pieces will be watching and monitoring your soil changes so that your application rates of the change of your practice is fully understood as well.

We've all made changes in our lives, and this is a major change for a farmer. But if you're fully informed and then you can look at what you need to do and change your equipment, how your soils will change, and you can monitor the changes. You can continue to make the decisions down the pathway to achieve that objective.

Internationally, the EU is really interested to see carbon farming taking a hold in Australia, and they wanna see farming on a trajectory to reduce their carbon emissions.

I think that the markets are going to start to dictate our carbon farming practices, and that will drive the change. So we might as well take a step forward and be farming down this direction ourselves. I think there's a significant risk if we don't adopt this carbon farming practice for Australia. A lot of farmers that we talked to are worried about the longevity of their soil and having to apply more and more fertiliser to get the same productivity so asset values are relying on longevity of productivity.

We've got a human health issue.

We can use less pesticide. We can use less herbicide. All of those things flow through to the quality of the food that we're all eating.

We're convinced that carbon farming is a win win. The improvement in the land value from carbon farming is gonna help the future generations as they come through. Carbon farming is also a win for the planet in that we're addressing sustainability.

Carbon farming is a win for the health of the consumer and that they're eating healthier food. So there's there's a triple win sitting in carbon farming. If you're interested to become a carbon farmer, our advice would be to talk to farmers that are in the game, talk to farmers that are carbon farming now, ask lots of questions, develop support groups amongst yourselves.

You can approach Tiverton. We're a demonstration farm. We're very happy to share all of our knowledge and help people on the carbon farming journey. It's critical to become very informed. Doing courses like this will help you gain your knowledge, talking to others, and being part of support groups. Being informed will give you the confidence to go forward because conventionally, there'll be a lot of people that'll try and talk you out of carbon farming.

There's no reason from our experience not to take that risk.

The opportunity is there for us. In 15 years, well, hopefully, we're not talking about carbon farming. We think it'll be an embedded practice because we're all interested in profitability, soil health, the health of the consumer, the products that we're selling to the world.

Hopefully, we're just talking about farming.

In this video (9:40 minutes), Matt Woods and Harry Youngman discuss soil carbon.

MATT WOODS: Hello. I'm Matt Woods and I'm with Harry Youngman to talk about soil carbon. Harry is a fourth generation farmer on the family property Ardgartan in southwest Victoria, where they run prime lambs, beef cattle, and some agroforestry.

Back in 2003, Harry began getting a bit disgruntled with certain aspects of the traditional farming system, and he decided to trial something different on Ardgartan. Harry, what did you trial and why did you do it?

HARRY YOUNGMAN: So, Matt, what prompted it was a sort of a diminishing return from the traditional systems of fertilising and stock ill-thrift and a demise in species composition.

So what we did was initially do a heap of plant tissue tests and showed up an imbalance of certain elements. And so then to rectify that, we first adopted a liquid fertiliser system using sulfates.

And then that was from sort of 2003 to 2008. And about 2008, we were introduced to compost, properly humidified compost.

And we embarked on a four year trial from 2008 to 2012, and we had some pretty spectacular results from that compost process.

We had control, zero tonnes and up to four tonnes per hectare. And then each year, we did quite a lot of microbiological testing and found some really impressive improvements.

WOODS: So I'm a farmer and or land manager, and I'm looking to improve my soil carbon. What's the sort of top line advice that Harry Youngman would give to a farmer or land manager looking to do that?

YOUNGMAN: So my advice would be to improve your knowledge of soil science or understanding of how the soil works in the first instance.

WOODS: What about, I know on Ardgartan, you did some trials before going into, heavily into certain aspects of this farming style. Should other farmers, do you think, put aside a portion of the property just to do a trial before they invest too heavily in this?

YOUNGMAN: I think we're blessed at the moment, where there's so much information and there's so many trials that have been done that people can research that, and understand what's happened. There's some great early adopters that are well down the path of the benefits of some of these processes.

However, there's nothing better than seeing is believing. And so if you do have a small scale trial or you wanna trial something like you may have done with your fertiliser trials on the farm, set something up. There's certain technology around that can allow you to do it on a paddock by paddock basis now, which is fantastic.

But there is no better education tool than seeing it for yourself. Now whether that's at a field day on your own farm, highly recommend undertaking some on farm trials.

WOODS: And can farmers expect to see instant results, or are they gonna need a bit of patience here to sort of keep going with the program?

YOUNGMAN: Farmers will see response in certain aspects very rapidly. For instance, they may see a reduction in red-legged earth mite infestations within 12 months.

However, they might not see soil carbon improvements for four to five years. So it's not a miracle that happens overnight, but there are definitely some canary in the coal mine type principles that you can pick up on and see change pretty quickly. You just gotta know what to anticipate and what to look for and most importantly, what to test for.

WOODS: Well, let's move on to testing then. What can they test for? How do you monitor this and know that your soil carbon is improving?

YOUNGMAN: So some of the early indicators that we've found to be reasonably good and reliable proxies for improvement in soil health. As I said at the start, after we did the four years of compost, we did some soil DNA testing, which showed dramatic improvements where extra compost had been put out.

Those tests have morphed, and you can have them done all around the world. We are using two tests based out of America now, which are testing DNA and improvements in soil microbial fungal populations and bacterial populations and the various diversity of microbes.

Bearing in mind that there's about eight billion microbes in a teaspoon of soil, you are not gonna see it with the naked eye. So you have to go to a DNA type of testing. As discussed, it's like the canary in the coal mine. So you are picking up well in advance DNA tests that you can't see with the human eye.

WOODS: Harry, what's some of the advice you would give farmers and land managers who are looking to improve their soil carbon and their soil health?

YOUNGMAN: One of the first things I'd look at doing is getting a piece of paper out and writing down all those things in the current production system that are frustrating you.

Maybe broadleaf weeds, maybe dags on sheep, it may be tensile strength in the fine wool merino operation, all those sorts of things that you just keep scratching your head about why is this happening.

Once you've done that, it'll help you synthesise down what are the important things that you might wanna try and address that no one's been able to help you with.

Secondly, you need to think through if you are going into a carbon project, how you're going to record everything that's going on on the farm in a concise and logical manner. It's a major part of any soil carbon project and it needs to be well thought out because the questions from the clean energy regulator and any auditors that might come onto your property will be pretty forensic.

WOODS: What about the knowledge that they're gonna need? Is that important to ensure maybe you're not talked out of your approach?

YOUNGMAN: Absolutely. Most of the time, people that have an agenda or a product to sell you will invariably talk down the benefits of self help, and they'll want to peddle you more of their product.

And so what my advice would be is just to empower yourself to be able to argue or discuss the point objectively and make it clear that you understand exactly what you're going to potentially replace their product with and be comfortable that it's going to work.

WOODS: Harry, it sounds like for farmers, it just makes business sense to improve the health of their soil. How did it work on Ardgartan in terms of the business outcomes?

YOUNGMAN: So what we've found on Ardgartan after benchmarking our whole operation, each enterprise for 20 plus years, is that we've been able to improve our output per hectare as measured by our dry sheep equivalent carrying capacity or DSE capacity. For instance, our district average is about 14 DSE per hectare.

WOODS: Harry, a farmer or landholder has gone down this path to improve their soil health. How are they gonna know if it's working or not?

YOUNGMAN: So some of the things that we saw were improved clover content, improved clover nodulation, less broad-leaf weeds, less dags on sheep. Cropping enterprises can probably look to see, less fungal infestations and other diseases. So I would think that healthier plants are more resilient and perform the way they are supposed to perform.

WOODS: Harry, why is it important that farmers and land managers get soil testing done, and and what sort of tests are out there that they can utilise?

YOUNGMAN: It's a bit like weighing your lambs or your cattle or measuring the yield of your crop. You need to measure so you can manage. There's chemical soil testing, there's biological soil testing, and there's soil structure soil testing. There's practical on farm soil testing you can do as simple as getting a shovel, digging some holes on your farm, and counting the worms that are in your shovel full.

That's a really good litmus test of how well your soil is functioning right now. The chemical soil tests will give you a really good picture on your macro elements, but also your carbon. You've probably already got carbon measurements going back many years if you've been testing for a long time. And you can sort of understand the cause and effect of different management practices that you may have overlaid across your property.

WOODS: Great. Harry, thank you very much for your time. Appreciate it.

YOUNGMAN: Thank you, Matt.

In this video (11:14 minutes), Matt Woods and Harry Youngman discuss vegetation carbon.

MATT WOODS: Hello. I'm Matt Woods, and I'm talking with Harry Youngman. Harry is a fourth generation farmer on the family property Ardgartan in southwest Victoria, where they run prime lambs, beef cattle, and some agroforestry.

Harry, you've planted a lot of woodlots and shelter belts on your property. Why did you choose to do this?

HARRY YOUNGMAN: Matt, I think my forebears possibly over cleared the property. And if you've ever been to the southwest of Victoria, you know, it's fairly inclement and harsh conditions. And so taking the lead from the Pastoral Veterinary Institute in Hamilton, and doctor Rod Bird back in the eighties, clearly demonstrated the benefits of shelter belts on farm. And I think he actually, deduced that you

could put between 12 and 15 percent shelter belts back on the farm before losing any productivity, such as the benefits of breaking up that wind and the benefits to livestock, etcetera, etcetera, which we've actually found on replicated trials across the farm. Our lambing percentages in our more sheltered paddocks are consistently between sort of seven percent to 13 percent better.

And, we're also finding good interception of recharge areas down to discharge areas and sort of combating some of the smaller salinity areas on the property. We've also observed the local indigenous bush around the farm, and we've sought to perpetuate that out in as groynes of biodiverse plantings.

The theory behind that is we create harbour for, birds and insects to help combat things like red-legged earth mite, and other parasites.

So they're the main benefits apart from the obvious beautification of the property.

WOODS: Planting trees contributes to storing carbon in vegetation, and that's all part of the carbon cycle.

Can you just give me a little brief overview of what the carbon cycle is?

YOUNGMAN: So we're blessed with a process called photosynthesis.

Plants will absorb CO₂ out of the atmosphere.

They will respire O₂, which keeps us alive, and they'll make complex organic compounds, which we see as biomass above the ground.

WOODS: So even if a farmer or land manager is not looking to get carbon credits by planting trees and getting vegetation carbon, what are some of the financial and business benefits they might see out of plantations?

YOUNGMAN: So on the plantation front, we've done both radiata pine and eucalyptus globulus, which is short rotation and long rotation plantations, both of which provided, very good return on investment and provided also blocks of capital for things like succession planning and debt repayment and also smoothing out or compensation of poor years.

WOODS: Harry, in terms of shelter belts then, what sort of advantages are farmers gonna see in their business by planting those?

YOUNGMAN: So in addition to the livestock benefits that we've talked about, the evapotranspiration is a massive factor that isn't talked about a lot.

And so breaking up that wind flow has been very, very beneficial to extending the growth of the season and maintaining or not losing so much moisture. The shelter belt's also at a micro scale on the farm. We're aiming to try and fit in with the macro sort of catchment management planning and thinking, and we've sought advice from our catchment management authority and indeed received grants for plantings to enhance that macro catchment benefit.

WOODS: And are they gonna see benefits in terms of predatory insects and things in those shelter belts?

YOUNGMAN: Yes. So our shelter belts consist of sort of shrubs, medium trees, and then even taller trees, which could potentially be used for, timber lot or, furniture grade material in the future. So in our plantations, we actually view the shrubs and small trees as being very, very valuable for the harbouring of important insects and bird life. An example of some of those insects might be a special wasp, which lives in the shrub called bursaria.

And that wasp is actually really effective in controlling red headed cockchafer.

The red headed cockchafer grub, which is the bane of the pasture farmer's life, is beyond control with conventional chemistry, but, interestingly enough, these wasps seem to be able to control them.

WOODS: And your own experience with that? You've seen that effectively happening on your property?

YOUNGMAN: Yes. So we have, had a dramatic reduction in the need for spraying red headed cockchafer and black headed cockchafer.

In fact, we don't spray for it anymore.

WOODS: So if I'm a farmer or land manager looking to get into this, what's Harry Youngman's kind of top level advice to start getting into it?

YOUNGMAN: So one of the key things is to develop and understand, what you want, and part of that process is developing a whole farm plan or a farm plan.

And that needs to respect things like waterways, roads, current roads and future roads, easements, title boundaries, council planting requirements, the right to harvest, and access to good public roads.

So shape and also the shape of plantations need to obviously consider the prevailing winds. Not only that, we need to understand that a whole farm plan is a microscopic or micro look at the broader, catchment and that in conjunction with people like your catchment management authority, you can look at the macro benefits of your plantings and how that may tie in with local indigenous bush or neighbours' plantings and indeed the waterways that feed into the major rivers in your catchment.

WOODS: So planting all those trees, you can add to your vegetation carbon. How are you gonna measure that vegetation carbon?

How are you gonna know how much you're sequestering?

YOUNGMAN: So the conventional forestry measurement systems are one method, and that's more in woodlot and having a random plot measurement system. The next level is, using the government provided FullCAM or full carbon accounting model, which actually predicts on a point and shoot basis how many tonnes of CO₂ equivalents you may be able to generate in your particular area using either mixed indigenous plantings or indeed Mallee plantings depending where you are and whether they're block or whether they're belt.

And so all those considerations need to be put in place. So they're all found in the Clean Energy Regulator's website.

WOODS: A farmer or land manager may not be looking for carbon credits now, but is there anything that they should think about or take into account? Because down the track, maybe they will be looking for carbon credits.

YOUNGMAN: So there are certain protocols that require a project to be registered before you actually plant. So you can't, today, retrospectively claim carbon credits on plantings that you have done. There's also some really good, technology coming along with not only from satellite, measures the volume of timber, but it also measures the health of the planting by picking up the gases that should be coming out, and it can give you a score out of 10 as to how healthy your biodiverse planting or your monoculture may or may not be.

So there's lots of, technology that's gonna help us do this for less cost and also do it more regularly.

WOODS: Harry, if a farmer or land manager wants to get into the carbon credits, is there any legal considerations they need to take into account?

YOUNGMAN: Matt, there are. There's, when you do go into a project, you need to consider, what encumbrance you are going to put on your land. And that may be determined by whether you elect for a 25 year permanence or a hundred year permanence, and it will vary. And also, in that process, they might like to consider the shape and style of project that they're going into, whether they're going to be employing a shelter belt model or a block planting model.

WOODS: It gets pretty complex, this stuff. Is there a place that farmers and land managers can go, you know, advisers or something that might be able to help them?

YOUNGMAN: Yeah. I strongly recommend getting not only legal advice, but also, try and find good carbon project developers that will help you work through not only the planning, the obligations, but also the potential pooling of those credits and effecting more lucrative sales.

WOODS: Harry, you mentioned FullCAM. What is that?

YOUNGMAN: So FullCAM stands for Full Carbon Accounting Modelling. And essentially, it's a point and shoot system, which was developed by the CSIRO and allows you to put in your coordinates and it will predict how many tonnes of CO₂ equivalents per hectare you can generate over a 25 year period.

WOODS: Harry, it all gets fairly complicated when you get into the carbon credits side of things. So where should land owners and farmers look for more information or maybe some expert advice?

YOUNGMAN: So we are blessed with a central body called the Carbon Market Institute, and all of the reputable, providers or service providers are actually subscribers to the agreed protocols of the CMI. And I would start with their website, and that will give you a list of, service providers.

In addition obviously, you need to canvass your trusted legal and accounting advisers as well.

WOODS: And maybe farmers could ask advice from other farmers that might have gone down this track?

YOUNGMAN: Absolutely. Yes. There's, so certainly, the Bush Telegraph is very, very appropriate in this situation where service providers have failed to provide the base service, farmers will be able to recognise that.

WOODS: Harry, that's been fascinating, and thank you very much for your time.

YOUNGMAN: Thanks, Matt.

Source: <https://www.agriculture.gov.au/agriculture-land/farm-food-drought/climatechange/carbon-farming-outreach-program/training-package/topic-4/6-case-study-expert-interviews>